

CSCI E-106:Assignment 4

Due Date: February 27, 2026 at 11:59 pm EST

Instructions

Students should submit their reports on Canvas. The report needs to clearly state what question is being solved, step-by-step walk-through solutions, and final answers clearly indicated. Please solve by hand where appropriate.

Please submit two files: (1) a R Markdown file (.Rmd extension) and (2) a PDF document, word, or html generated using knitr for the .Rmd file submitted in (1) where appropriate. Please, use RStudio Cloud for your solutions.

Problem 1

Refer to Sales growth Data. (30 points, 10 points each)

a-) Divide the range of the predictor variable (coded years) into five bands of width 2.0, as follows: Band 1 ranges from $X = -.5$ to $X = 1.5$; band 2 ranges from $X = 1.5$ to $X = 3.5$; and so on. Determine the median value of X and the median value of Y in each band and develop the band smooth by connecting the five pairs of medians by straight lines on a scatter plot of the data. Does the band smooth suggest that the regression relation is linear? Discuss.

b-) Create a series of seven overlapping neighborhoods of width 3.0 beginning at $X = -.5$. The first neighborhood will range from $X = -.5$ to $X = 2.5$; the second neighborhood will range from $X = .5$ to $X = 3.5$; and so on. For each of the seven overlapping neighborhoods, fit a linear regression function and obtain the fitted value $\hat{Y}_c$ at the center X_c of the neighborhood. Develop a simplified version of the lowess smooth by connecting the seven $(X_c, \hat{Y}_c)$ pairs by straight lines on a scatter plot of the data.

c-) Obtain the 95 percent confidence band for the true regression line and plot it on the plot prepared in part (b). Does the simplified lowess smooth fall entirely within the confidence band for the regression line? What does this tell you about the appropriateness of the linear regression function?

Problem 2

Using the ozone data under faraway library, if you are using the R Studio, please use the command below to download the data:

```
library(faraway)
data(ozone, package="faraway")
```

If you are using Posit R Cloud instead of RStudio Desktop, please use the command below to download the data:

```
tmp <- tempfile(fileext = ".rda")
download.file(
  "https://raw.githubusercontent.com/cran/faraway/master/data/ozone.rda",
  destfile = tmp,
  mode = "wb"
)
load(tmp)
head(ozone)
```

```
##   O3   vh wind humidity temp  ibh dpg ibt vis doy
## 1  3 5710    4      28   40 2693 -25  87 250  33
## 2  5 5700    3      37   45  590 -24 128 100  34
## 3  5 5760    3      51   54 1450  25 139  60  35
## 4  6 5720    4      69   35 1568  15 121  60  36
## 5  4 5790    6      19   45 2631 -33 123 100  37
## 6  4 5790    3      25   55  554 -28 182 250  38
```

Use O3 as the response and temp, humidity and ibh as predictors to answer the questions below. (40 points)

- Create train and test data sets. Use 70% of the data for train data set and use remaining data (30% of data) for test data set (use `set.seed(1023)`). (5 points)
- Build a regression model to predict O3 on the train data set. Write down the regression model and comment on the model performance. (5 points)
- Visually by using the graphs, comments on regression model assumptions. (5 points)
- Test the model performance on the test data and comment of model stability and performance. (5 points)
- Perform Breusch-Pagan Test, write down null and alternative hypotheses and your conclusion. (5 points)
- Use the Box-Cox method to determine the best transformation on the response. if transformation is needed, perform the transformation and rebuild the model on the train data and check the model stability on the test data set. (15 points)

Problem 3

Using the teengamb data under faraway library, if you are using the R Studio, please use the command below to download the data:

```
library(faraway)
data(teengamb, package="faraway")
```

If you are using Posit R Cloud instead of RStudio Desktop, please use the command below to download the data:

```
load(url("https://raw.githubusercontent.com/cran/faraway/master/data/teengamb.rda"))
head(teengamb)
```

```
##   sex status income verbal gamble
## 1   1     51    2.00      8     0.0
## 2   1     28    2.50      8     0.0
## 3   1     37    2.00      6     0.0
## 4   1     28    7.00      4     7.3
## 5   1     65    2.00      8    19.6
## 6   1     61    3.47      6     0.1
```

- a-) Fit a model with gamble as the response and the other variables as predictors, and Visually by using the graphs, comments on regression model and assumptions. (10 points total).
- b-) Predict the data(teengamb) amount that a male with average (given these data) status, income and verbal score would gamble along with an appropriate 95% CI (5 points).
- c-) Repeat the prediction for a male with maximal values (for this data) of status, income and verbal score. Which CI is wider and why is this result expected? (5 points)
- d-) Fit a model with sqrt (gamble) as the response but with the same predictors. Now predict the response and give a 95% prediction interval for the individual in (a). Take care to give your answer in the original units of the response (5 points).
- e-) Repeat the prediction for the model in (d) for a female with status=20, income=1, verbal = 10. Comment on the credibility of the result (5 points).